Prime-Pattern Optical Resonator & Aperiodic AR System

Inventor: Branden Lee Friend
Date: 2025-10-04

1. Overview

This document describes a combined aperiodic anti-reflection (AR) coating and photonic resonator system based on logarithmic-prime scaling of layer thicknesses or optical path lengths. The design achieves deterministic broadband destructive interference and tunable spectral resonances that map directly to the Riemann ζ-function zeros via a physical operator correspondence.

Parameter	Specification
Platform	SiN on SiO■ (1550 nm C-band)
Core thickness	300 nm ± 5 nm
Propagation loss	≤ 0.2 dB/cm
Heater efficiency	25 mW per 2π phase shift
AR coating	Prime-pattern TiO■/SiO■ multilayer (16 layers)
Die size	35 mm × 25 mm
Package	24-pin butterfly / fiber array (8×8)

2. Layer Stack (AR Coating)

The AR coating employs an aperiodic quarter-wave structure where each layer thickness t■ = (λ■ / 4n■) × (log p / log 2). This deterministic prime sequence yields reflection suppression over 1500–1600 nm with R ≤ –30 dB.

Layer	Material	n (1550 nm)	t [nm]
1	TiO■	2.35	165
2	SiO■	1.44	285
3	TiO■	2.35	180
4	SiO■	1.44	295
5	TiO■	2.35	205
6	SiO■	1.44	320
7	TiO■	2.35	220
8	SiO■	1.44	340

3. Quantum-Tube Prime Resonator

The photonic resonator consists of a planar array of SiN waveguides where each prime p corresponds to a path length ■■ = α log p. Couplers implement the operators T■ and T■*, with phase and amplitude weights p^(–σ) and e^(–it log p). Sweeping the global phase parameter t reproduces the spectral signature of ζ(s) along the critical line Re(s)=1/2.

Key dimensions and components:

Component	Specification
Waveguide core	SiN 300 nm × 1.2 µm
Effective index	n_eff ≈ 1.75
Prime path lengths	■■=3.4 mm to ■■■≈17 mm
Phase shifters	TiN heaters (25 mW / 2π)
Variable couplers	Thermo-optic VDC, ±2% tuning
Control lines	26 total (phases + amplitudes)

4. Acoustic Demonstrator

A low-cost acoustic analog can be constructed using aluminum or 3D-printed tubes with lengths L■ = α log p (α = 0.33 m). Couplers form T-junction manifolds, and phase control is achieved with adjustable stubs. Swept-frequency analysis reveals minima matching finite-P ζ-zero approximants.

5. Patent Summary

Independent claims cover any optical or acoustic system implementing a log-prime layer or path-length rule to achieve deterministic broadband destructive interference or ζ-correlated resonances. Trade secrets include optimization algorithms, scaling coefficients, and calibration methods.

6. Commercialization

Prime-Grade AR™ and Prime Resonator™ will be licensed to photonics OEMs and research labs. Use cases include LiDAR, telecom windows, and metrology. Expected royalties: \$0.005–\$0.01 per cm^2 coating; \$50–\$200 per PIC demo unit.